Adaptation Effects in Two and Three Layer Networks

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Network

$$f(x) = \sum_{i=1}^{N} a_i \, \mathrm{erf}(w_i \cdot x), \qquad x \in \mathbb{R}^d$$

$$D = \{(x_\mu, y_\mu)\}_{\mu=1}^{P}$$

Targets and feature integrals

$$
\begin{aligned}
d\mu_x &= p(x)\,dx, \\
y(x) &= \mathrm{He}_1(w_* \cdot x) + \varepsilon\,\mathrm{He}_3(w_* \cdot x), \qquad \varphi_m(x) = \mathrm{He}_m(w_* \cdot x), \\
y_m &= \langle y, \varphi_m \rangle_x = \int d\mu_x\, y(x)\,\varphi_m(x), \\
c_m(w) &= \langle \mathrm{erf}(w \cdot x), \varphi_m \rangle_x = \int d\mu_x\, \mathrm{erf}(w \cdot x)\,\varphi_m(x).
\end{aligned}
$$

Bayesian action

$$p(a, W \mid D) \propto e^{-S[a,W]}$$

$$S[a,W] = \frac{1}{2\sigma_A^2}\|a\|^2 + \frac{d}{2\sigma_w^2}\|W\|^2 + \frac{P\chi}{2\kappa}\int d\mu_x \left(f(x) - y(x)\right)^2$$

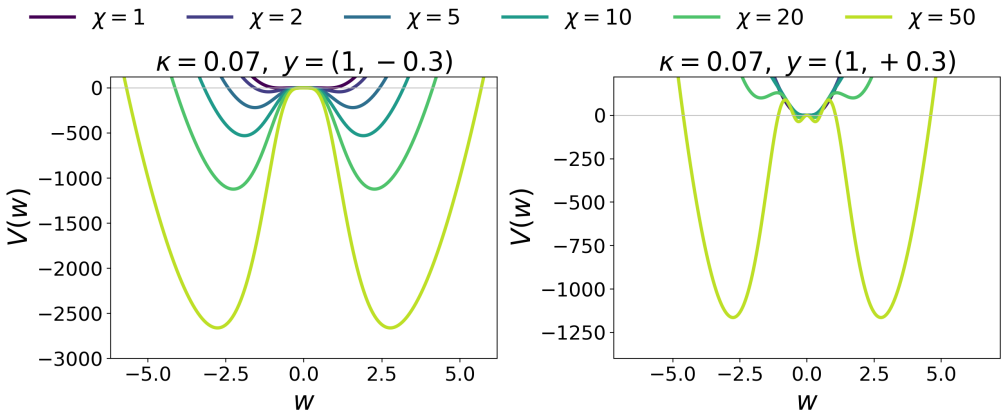
κ	ridge
P	dataset size
N	width
d	input dimension
$\sigma_w^2,\ \sigma_A^2$	prior variances
χ	loss scale

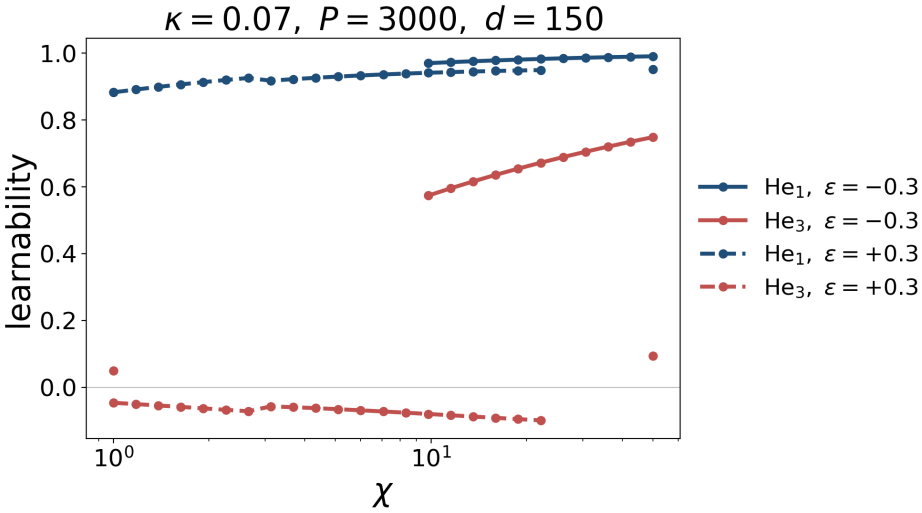
Phenomenology of Neural Adaptation

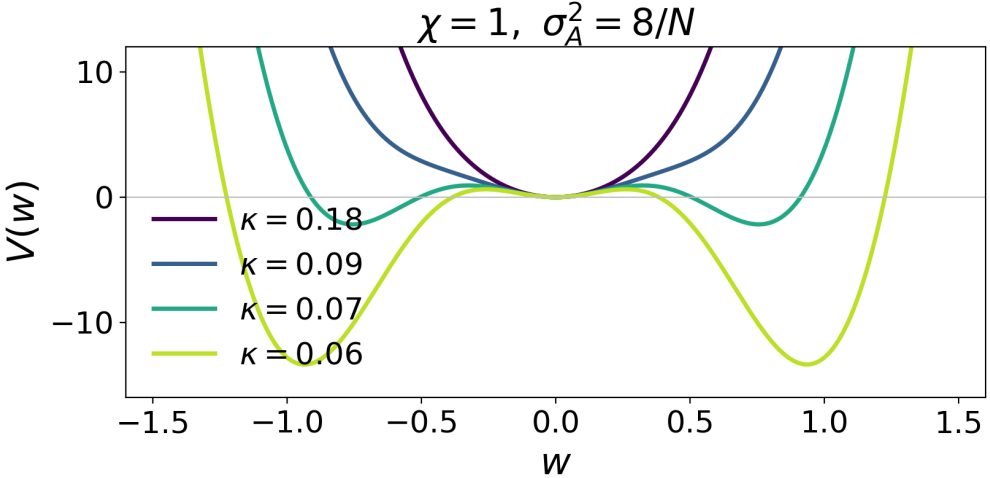
$$S[w \cdot w_*] = \frac{d}{2\sigma_w^2}\left[(w \cdot w_*)^2 - \frac{\sigma_w^2 \sigma_A^2 \chi}{d\,N}\Big(t(\tilde{L})\,(w \cdot w_*) + g(\tilde{L})\,(w \cdot w_*)^3 + u(\tilde{L})\,(w \cdot w_*)^5\Big)^2\right]$$

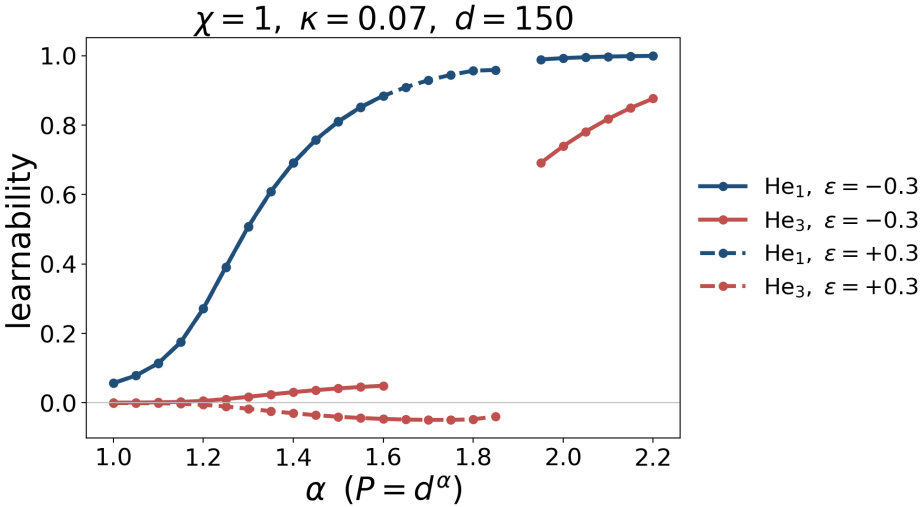
with discrepancy

$$\tilde{L}_m = \frac{P}{\kappa}\left\langle y - f,\ \varphi_m \right\rangle_x = \frac{P}{\kappa}\int d\mu_x \left(y(x) - f(x)\right)\varphi_m(x).$$


$\chi = 1$
$\chi = 2$
$\chi = 5$
$\chi = 10$
$\chi = 20$
$\chi = 50$
$\kappa = 0.07,\ y = (1, -0.3)$
$\kappa = 0.07,\ y = (1, +0.3)$
$V(w)$
w
0
−500
−1000
−1500
−2000
−2500
−3000
−5.0
−2.5
0.0
2.5
5.0
−250
−750
−1250


$\kappa = 0.07,\ P = 3000,\ d = 150$
learnability
1.0
0.8
0.6
0.4
0.2
0.0
10^0
10^1
χ
$\mathrm{He}_1,\ \varepsilon = -0.3$
$\mathrm{He}_3,\ \varepsilon = -0.3$
$\mathrm{He}_1,\ \varepsilon = +0.3$
$\mathrm{He}_3,\ \varepsilon = +0.3$


$\chi = 1,\ \sigma_A^2 = 8/N$
10
0
−10
$V(w)$
$\kappa = 0.18$
$\kappa = 0.09$
$\kappa = 0.07$
$\kappa = 0.06$
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
w


$\chi = 1,\ \kappa = 0.07,\ d = 150$
learnability
1.0
0.8
0.6
0.4
0.2
0.0
1.0
1.2
1.4
1.6
1.8
2.0
2.2
$\alpha\ \ (P = d^{\alpha})$
$\mathrm{He}_1,\ \varepsilon = -0.3$
$\mathrm{He}_3,\ \varepsilon = -0.3$
$\mathrm{He}_1,\ \varepsilon = +0.3$
$\mathrm{He}_3,\ \varepsilon = +0.3$

Kernels of the preactivations

Each preactivation layer carries a kernel K. For the erf read-in,

$$K = \sigma_A^2\, J, \qquad J(x, x') = \mathbb{E}\big[\mathrm{erf}(w \cdot x)\ \mathrm{erf}(w \cdot x')\big].$$

$$R = K + \frac{\kappa}{P}\, I, \qquad L = K\, R^{-1}, \qquad f = L\, y, \qquad y - f = \frac{\kappa}{P}\, R^{-1} y.$$

Hence $\tilde{L} = R^{-1} y$ in the φ_m basis. A three-layer net has the same object on the next preactivation.

Kernel Ridge Regression in Readout Layer

The readout learns a feature map: $\phi_i(x) = \mathrm{erf}(w_i \cdot x)$, with

$$\sum_i a_i \ \mathrm{erf}(w_i \cdot x) = \big(KR^{-1}y\big)(x),$$
$$= \big(L\,y\big)(x).$$

Exact fit in the eigenbasis

In the eigenbasis of K, with $K\varphi_m = \lambda_m \varphi_m$,

$$\big(KR^{-1}y\big)(x) = \sum_m \frac{\lambda_m}{\lambda_m + \kappa/P}\, y_m\, \varphi_m(x).$$

The same basis expands the activation,

$$\operatorname{erf}(w \cdot x) = \sum_m c_m(w)\, \varphi_m(x).$$

The readout therefore meets $KR^{-1}y$ mode by mode,

$$\sum_i a_i\, c_m(w_i) = \frac{\lambda_m}{\lambda_m + \kappa/P}\, y_m.$$

Feature learning: GFL and grokking

What the network expresses

Sample complexity

Barrier height and time

Linear adaptation

Grokking

Inner well at larger χ

Cubic learnability: NNGP and rich